

Sprint 6 Report

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Overview

My goal for Sprint 6 was to finish the Chart Editor system in my 3D rhythm game, which allows players to create charts of notes for song audios. This goal falls under two acceptable forms of progress from the assignment:

1. “Added features and enhancements”
2. “User experience enhancements”

To complete the chart editor, I planned to touch up the UI features I had, and mainly fix the logic behind the chart editor, including playback, note placement, adding shortcuts, event creation, and audio processing.

More information on my project can be found in my GitHub repo:

<https://github.com/rk3026/RhythmGame>

Product Documentation

- **User documentation:**
 - **This is a Windows Release only**
 - Download the game from my Github Repo releases:
<https://github.com/rk3026/RhythmGame/releases/tag/v0.1.0-alpha>
 - Unzip the file and run RhythmGame.exe or RhythmGame.console.exe
- **Feature overview:**
 - Profile Creation, Editing, Export, Import, and Delete
 - Achievement and Level system
 - Quickplay
 - Select a song
 - Gameplay of the song
 - Results
 - Chart editor
 - Import audio
 - Wavelength detection
 - Shortcuts to control placing notes and playback
 - BPM detection

- Settings
 - Volume
 - Key bindings
- **Known limitations:**
 - Copyright issues with songs. I utilize several well-known songs that have Copyright protection, so in future releases these must be removed to avoid Copyright infringement.

Version Control Information

- **Repository link:** <https://github.com/rk3026/RhythmGame>
- **Sprint 5 commit ID:**
<https://github.com/rk3026/RhythmGame/commit/52072f3eeea615a7052b09df9e6a350d56402196>
- **Sprint 6 commit ID:**
<https://github.com/rk3026/RhythmGame/commit/2d7f4d36e4ab677a59b24b3c59675408258ed75f>
- **Separate branch:** sprint6 - <https://github.com/rk3026/RhythmGame/tree/sprint6>

Agentic Coding Workflow Documentation

- **Detailed workflow description:**
 - 3 agents:
 - Visuals agent
 - GitHub Copilot
 - Claude Sonnet 4.5
 - Focused on .tsch (scene) files, node structure, node references, and UI/UX.
 - Logic agent
 - GitHub Copilot
 - Claude Sonnet 4.5
 - Focused on .gd files, the logic of how the game works, including note processing, sound effects, events, etc.
 - Ideas agent
 - ChatGPT, Claude, and Gemini websites
 - Focused on generating ideas for the game, such as new gameplay features and animations.

- Handoffs of documents between agents was done manually. I act as the overseer of the agents work. If they make a mistake or poor implementation, I am able to catch it and step in.
- **Tool configuration:**
 - To create the GitHub Copilot agents, I utilized `.github/.chatmodes` markdown files. These can be used by the Copilot Chat extension as agent descriptions. The LLMs will follow instructions in these files when the agent mode is set to that chatmode.
 - I also used `.github/instructions.md` as a way to give context to my agents. The agents will always be given the contents of `instructions.md`, so I put instructions and more details on how Godot works for them to better work with the engine.
 - I utilized the Godot MCP server (<https://github.com/Coding-Solo/godot-mcp>) for the agents to be able to access features of the game engine. This made it easier for the agents to get output from the engine and view scene structure, etc.
- **Workflow evidence:**
 - Logs
 - `.github/chart-editor-log1.md`
 - `.github/chart-editor-log2.md`
 - `.github/copilot-logs.md`
- **Reflection:** My AI workflow mostly stayed the same from Sprint 5 to Sprint 6. I found this workflow was the most efficient. Some models were added to GitHub Copilot during my Sprint 6, including Claude Opus 4.5, GPT-5.1-Codex-Max, GPT-5.1-Codex-Mini, and GPT-5.2. I found the new Opus 4.5 model helpful for debugging some issues I had in Sprint 6 as I implemented my new features.

Sprint 6 Summary

- **What you accomplished:**
 - Finishing up the chart editor. This involved adding more note placements tools (sustain notes), keyboard shortcuts for placing notes and controlling playback, bpm and wavelength detection of song audio, and bug fixes of playback in the chart editor system.
 - Building the game and adding a release to GitHub to allow people to download and play the game.
- **Connection to Sprint 5 goals:**

- After sprint 5, I identified how the chart editor was unfinished. My goal for sprint 6 was to finish the chart editor.
- **Key challenges overcome:**
 - Build issues (inconsistencies between and build). The game did directory scanning before, which did not work when the game was built. I switched to making manifests for the songs.
 - Chart editor bugs. Playback had issues, such as notes skipping around, and hold notes not being properly placed during playback.
- **Demo video link:** https://youtu.be/Lw_JIye0pg4